

Homework 1

Hanvesh Aduru

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1. Factorization and Divisibility

(a)

Factor

$$2^{15} - 1 = 32,767.$$

Using the difference of cubes,

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2),$$

we get

$$2^{15} - 1 = (2^5)^3 - 1^3.$$

Therefore,

$$(2^5 - 1)(2^{10} + 2^5 + 1).$$

Since

$$2^5 - 1 = 31$$

and

$$2^{10} + 2^5 + 1 = 1024 + 32 + 1 = 1057,$$

we have

$$\boxed{32,767 = 31 \cdot 1057}.$$

(b)

Since

$$2^{15} - 1 = 32,767$$

divides

$$2^{32,767} - 1,$$

we can choose

$$\boxed{x = 32,767}.$$

This satisfies

$$1 < x < 2^{32,767} - 1.$$

2. Conjectures

Consider the following values:

n	$3^n - 1$	Prime?	$3^n - 2^n$	Prime?
1	2	Yes	1	No
2	8	No	5	Yes
3	26	No	19	Yes
4	80	No	65	No
5	242	No	211	Yes
6	728	No	665	No

A conjecture is that for $n > 1$,

$$3^n - 1$$

is never prime. This is because 3^n is odd, so $3^n - 1$ is even and greater than 2.

Also, $3^n - 2^n$ can be prime for some values of n , but not for every value of n .

If n is composite, then $3^n - 2^n$ is also composite because expressions of the form

$$a^n - b^n$$

can be factored when n is composite.

3. Finding a New Prime

The method is to multiply the given primes together and add 1.

(a)

For the primes 2, 3, 5, and 7:

$$2 \cdot 3 \cdot 5 \cdot 7 + 1 = 211.$$

Since 211 is prime and is not in the original list,

$$\boxed{211}$$

is the required prime.

(b)

For 2, 5, and 11:

$$2 \cdot 5 \cdot 11 + 1 = 111.$$

However,

$$111 = 3 \cdot 37,$$

so 111 is not prime.

Including 3 and trying again gives

$$2 \cdot 5 \cdot 11 \cdot 3 + 1 = 331.$$

Since 331 is prime,

$$\boxed{331}$$

is a prime different from 2, 5, and 11.

4. Five Consecutive Non-Prime Integers

Five consecutive integers that are not prime are

$$\boxed{24, 25, 26, 27, 28}.$$

Indeed,

$$24 = 2 \cdot 12,$$

$$25 = 5 \cdot 5,$$

$$26 = 2 \cdot 13,$$

$$27 = 3 \cdot 9,$$

and

$$28 = 2 \cdot 14.$$

Therefore none of these five integers is prime.

5. Perfect Numbers

The first perfect numbers are

$$6, \quad 28, \quad 496, \quad 8128.$$

Therefore, the two additional perfect numbers are

$$\boxed{496 \text{ and } 8128}.$$

For 496,

$$1 + 2 + 4 + 8 + 16 + 31 + 62 + 124 + 248 = 496.$$

For 8128,

$$1 + 2 + 4 + 8 + 16 + 32 + 64 + 127 + 254 + 508 + 1016 + 2032 + 4064 = 8128.$$

6. Triplet Primes

The prime triplets in which each pair of adjacent numbers differs by 2 are

$$(3, 5, 7)$$

and

$$(5, 7, 11).$$

These are the only two such triplets.

7. Amicable Numbers

We show that $(220, 284)$ is an amicable pair.

The positive divisors of 220 that are smaller than 220 are

$$1, 2, 4, 5, 10, 11, 20, 22, 44, 55, 110.$$

Their sum is

$$1 + 2 + 4 + 5 + 10 + 11 + 20 + 22 + 44 + 55 + 110 = 284.$$

Now consider 284. Its positive divisors smaller than 284 are

$$1, 2, 4, 71, 142.$$

Their sum is

$$1 + 2 + 4 + 71 + 142 = 220.$$

Thus, the sum of the proper divisors of 220 is 284, and the sum of the proper divisors of 284 is 220.

Therefore,

$$(220, 284) \text{ is an amicable pair.}$$